1.7.6 Proof by Contraposition

Proofs of the theorems of the type where they do not start with the premises and end with the conclusion, are called $\boxed{\text{indirect proofs}}$.

An extremely useful type of indirect proof is known as $\boxed{\text{proof by contraposition}}$.

Proofs by contraposition make use of the fact that the conditional statement $p \to q$ is equivalent to its contrapositive, $\neg q \to \neg p$. This means that the conditional statement $p \to q$ can be proved by showing that its contrapositive, $\neg q \to \neg p$, is true. In a proof by contraposition of $p \to q$, we take $\neg q$ as a premise, and using axioms, definitions and previously proven theorems, together with rules of inference, we show that $\neg p$ must follow.

Example 3: Prove that if n is an integer and $3n+2$ is odd, then n is odd.

Soln: We will give a proof by contraposition.

Let p: $3n+2$ is odd q: n is odd

$\neg p$: $3n+2$ is even $\neg q$: n is even.

Let, n be an arbitrary even integer. By Definition 1, there exists an integer K such that $n = 2K$. We know $3K+1$ is an integer

$\therefore 3n+2 = 3(2K)+2 = 2(3K+1)$
$= 2K'$, where $K' = 3K+1$

By Definition 1, $3n+2$ is even. $\square$

Example 4: Prove that if $n = ab$, where a and b are positive integers, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$.

$Q(n)$: $a \leq \sqrt{n}$ $R(n)$: $b \leq \sqrt{n}$.

Soln: $P(n)$: $n = ab$

Domain: The set of positive integers

$\forall n (P(n) \to Q(n) \vee R(n))$. ~~The contrapositive is~~

We will have to prove $Q(n) \vee R(n)$ if $P(n)$ is True.